NU Information Exchange System - Part 1

Application Section

1. Application Design

The NU-Information Exchange System implements a client-server architecture where a Central Server acts as the communication hub for multiple campus clients. The system uses a hybrid TCP/UDP approach for different communication needs.

Central Server (central_server.cpp):

- Manages multiple campus client connections using multi-threading (one thread per client)
- Authenticates clients using hardcoded credentials (e.g., Lahore: NU-LHR-123)
- Routes TCP messages between campuses
- Receives UDP heartbeat packets every 10 seconds from clients
- Provides an admin console for monitoring connections and broadcasting announcements

Campus Client (campus_client.cpp):

- Connects to the server via TCP for authentication and messaging
- Sends UDP heartbeat packets every 10 seconds to indicate online status
- Provides a console menu for users to send/receive messages
- Runs four concurrent threads: main menu, TCP receiver, UDP heartbeat sender, and UDP broadcast listener

2. Custom Protocol

All messages use a pipe-delimited format for easy parsing:

Authentication (Client → Server):

AUTH|Campus:Lahore|Pass:NU-LHR-123|UDPPORT:6001\n

Authentication Response (Server → Client):

AUTH_OK|Campus:Lahore\n

AUTH_FAIL|ERR:WrongPassword\n

Inter-Campus Message (Client → Server → Client):

MSG|FROM:Lahore|TO:Karachi|DEPT:Academics|BODY:Meeting at 3pm\n

Heartbeat (Client → Server via UDP):

HEART|Campus:Lahore|TS:1702012345\n

Broadcast (Server → All Clients via UDP):

BCAST|TS:1702012345|BODY:System maintenance tonight\n

Protocol Design Rationale:

- TCP is used for authentication and inter-campus messages requiring reliable delivery
- UDP is used for heartbeats (loss of one packet is acceptable) and broadcasts (low overhead for multiple recipients)

3. Compilation Instructions

Compile both programs using g++ with pthread support:

bash

g++ central_server.cpp -pthread -o central_server

g++ campus_client.cpp -pthread -o campus_client

Note: The -pthread flag is required for multi-threading support.

4. Execution Instructions

Step 1: Start the Central Server

bash

./central_server 5000 5001

- Port 5000: TCP port for client connections
- Port 5001: UDP port for heartbeats

Step 2: Start Campus Clients (in separate terminals)

Lahore Campus:

bash

./campus_client 127.0.0.1 5000 5001 Lahore 6001

Karachi Campus:

bash

./campus_client 127.0.0.1 5000 5001 Karachi 6002

Islamabad Campus:

bash

./campus_client 127.0.0.1 5000 5001 Islamabad 6003

Parameters:

- 127.0.0.1: Server IP (use actual IP for remote connections)
- 5000: Server TCP port
- 5001: Server UDP port
- Lahore/Karachi/Islamabad: Campus name
- 6001/6002/6003: Client's UDP port for receiving broadcasts

Step 3: Authenticate When prompted, enter the password for your campus:

- Lahore: NU-LHR-123
- Karachi: NU-KHI-123
- Islamabad: NU-ISB-123

5. Admin Console Commands

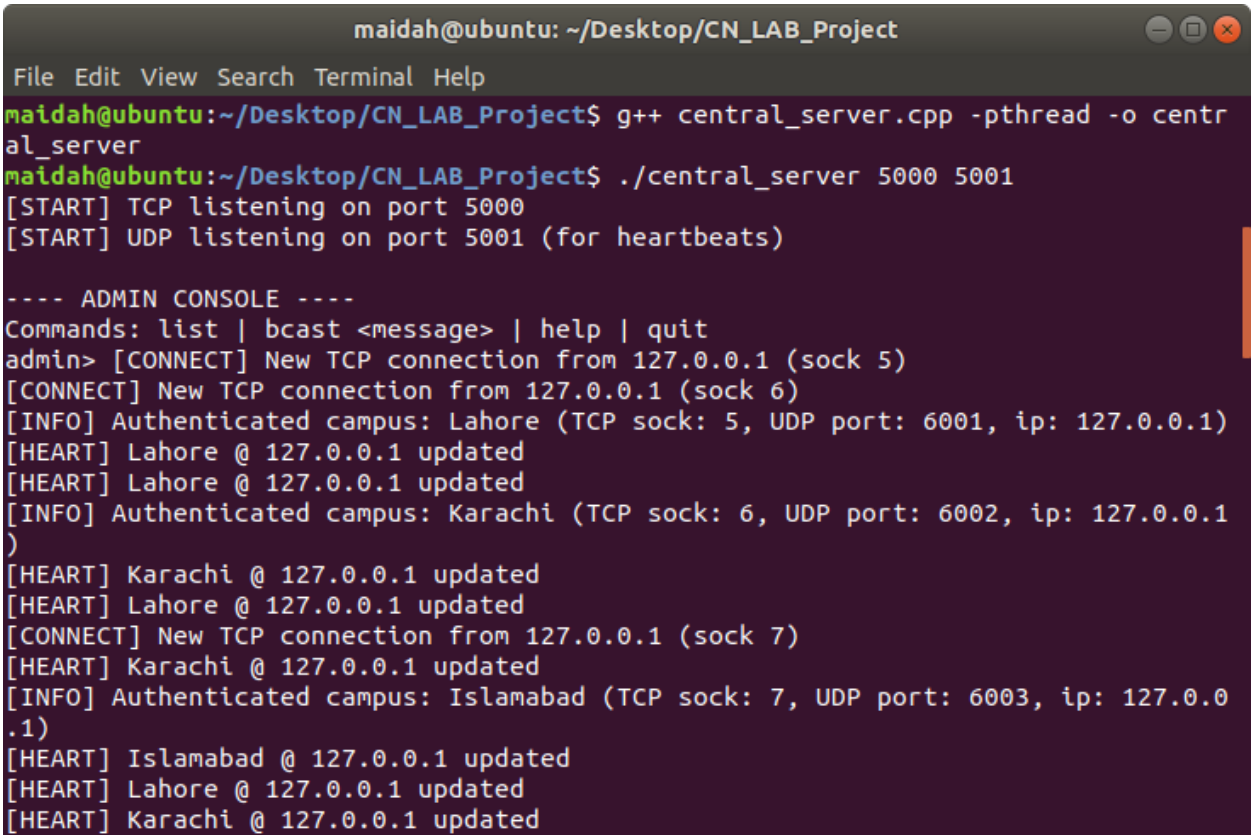
The server provides an interactive admin console with the following commands:

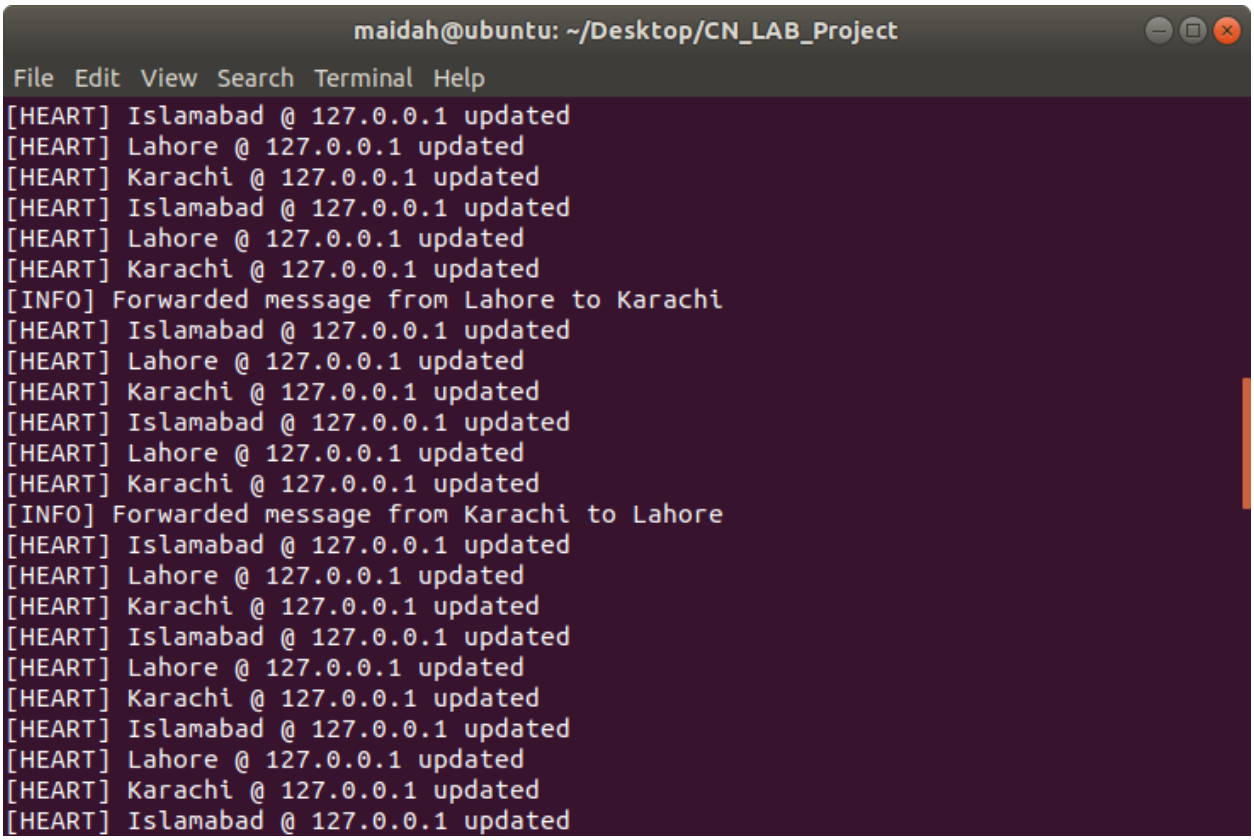
- list - Display all connected campuses with their last heartbeat time
- bcast <message> - Send a UDP broadcast announcement to all connected campuses
- help - Show available commands
- quit - Shutdown the server

6. Sample Output

Central_Server.cpp

a) **Connecting 3 campus_client and passing messages:**


maidah@ubuntu: ~/Desktop/CN_LAB_Project
File Edit View Search Terminal Help
maidah@ubuntu:~/Desktop/CN_LAB_Project$ g++ central_server.cpp -pthread -o central_server
maidah@ubuntu:~/Desktop/CN_LAB_Project$ ./central_server 5000 5001
[START] TCP listening on port 5000
[START] UDP listening on port 5001 (for heartbeats)
---- ADMIN CONSOLE ----
Commands: list | bcast <message> | help | quit
admin> [CONNECT] New TCP connection from 127.0.0.1 (sock 5)
[CONNECT] New TCP connection from 127.0.0.1 (sock 6)
[INFO] Authenticated campus: Lahore (TCP sock: 5, UDP port: 6001, ip: 127.0.0.1)
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[INFO] Authenticated campus: Karachi (TCP sock: 6, UDP port: 6002, ip: 127.0.0.1)
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[CONNECT] New TCP connection from 127.0.0.1 (sock 7)
[HEART] Karachi @ 127.0.0.1 updated
[INFO] Authenticated campus: Islamabad (TCP sock: 7, UDP port: 6003, ip: 127.0.0.1)
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated


maidah@ubuntu: ~/Desktop/CN_LAB_Project
File Edit View Search Terminal Help
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[INFO] Forwarded message from Lahore to Karachi
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[INFO] Forwarded message from Karachi to Lahore
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated

```
maidah@ubuntu: ~/Desktop/CN_LAB_Project
File  Edit  View  Search  Terminal  Help
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[INFO] Forwarded message from Islamabad to Karachi
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[INFO] Forwarded message from Karachi to Islamabad
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
[HEART] Lahore @ 127.0.0.1 updated
[INFO] Campus Karachi disconnected (sock 6)
[INFO] Removed client record for Karachi
[INFO] Campus Islamabad disconnected (sock 7)
[INFO] Removed client record for Islamabad
[INFO] Campus Lahore disconnected (sock 5)
[INFO] Removed client record for Lahore
```

b) Showing the connected active clients from admin(central_server) side:

```
admin> [HEART] Lahore @ 127.0.0.1 updated
list[HEART] Karachi @ 127.0.0.1 updated

Connected campuses:
 - Islamabad | TCP: 6 | ip: 127.0.0.1 | udp_port: 6003 | last hb(s) ago: 8
 - Karachi | TCP: 5 | ip: 127.0.0.1 | udp_port: 6002 | last hb(s) ago: 0
 - Lahore | TCP: 8 | ip: 127.0.0.1 | udp_port: 6001 | last hb(s) ago: 2
admin> [HEART] Islamabad @ 127.0.0.1 updated
```

c) If we quite from admin side, the clients automatically gets disconnected:

Central_server.cpp

```
admin> [HEART] Lahore @ 127.0.0.1 updated
[HEART] Karachi @ 127.0.0.1 updated
[HEART] Islamabad @ 127.0.0.1 updated
quit
Shutting down server (admin requested quit)...
```

Campus_client.cpp

```
[INFO] Connected to server.
Enter password for campus 'Lahore': NU-LHR-123
[INFO] Authentication successful.
---- Campus Client: Lahore ----
menu> [ERROR] Server disconnected or TCP recv error. Exiting...
```

```
maidah@ubuntu:~/Desktop/CN_LAB_Project$ ./campus_client 127.0.0.1 5000 5001 Karachi 6002
[INFO] Connected to server.
Enter password for campus 'Karachi': NU-KHI-123
---- Campus Client: Karachi ----
menu> [INFO] Authentication successful.
[ERROR] Server disconnected or TCP recv error. Exiting...
```

```
maidah@ubuntu:~/Desktop/CN_LAB_Project$ ./campus_client 127.0.0.1 5000 5001 Islamabad 6003
[INFO] Connected to server.
Enter password for campus 'Islamabad': NU-ISB-123
---- Campus Client: Islamabad ----
menu> [INFO] Authentication successful.
[ERROR] Server disconnected or TCP recv error. Exiting...
```

Campus_client.cpp

Lahore campus:

```
maidah@ubuntu:~/Desktop/CN_LAB_Project$ g++ campus_client.cpp -pthread -o campus_client
maidah@ubuntu:~/Desktop/CN_LAB_Project$ ./campus_client 127.0.0.1 5000 5001 Lahore 6001
[INFO] Connected to server.
Enter password for campus 'Lahore': NU-LHR-123
---- Campus Client: Lahore ----
menu> [INFO] Authentication successful.
1
Target Campus: Karachi
Department: Academics
Message: Send the deaprtments list
[SENT]
menu>
[INCOMING] MSG|FROM:Karachi|TO:Lahore|DEPT:Academics|BODY:Sent via email
menu> ^C
```

Karachi Campus:

```
maidah@ubuntu:~/Desktop/CN_LAB_Project$ g++ campus_client.cpp -pthread -o campus_client
maidah@ubuntu:~/Desktop/CN_LAB_Project$ ./campus_client 127.0.0.1 5000 5001 Karachi 6002
[INFO] Connected to server.
Enter password for campus 'Karachi': NU-KHI-123
---- Campus Client: Karachi ----
menu> [INFO] Authentication successful.

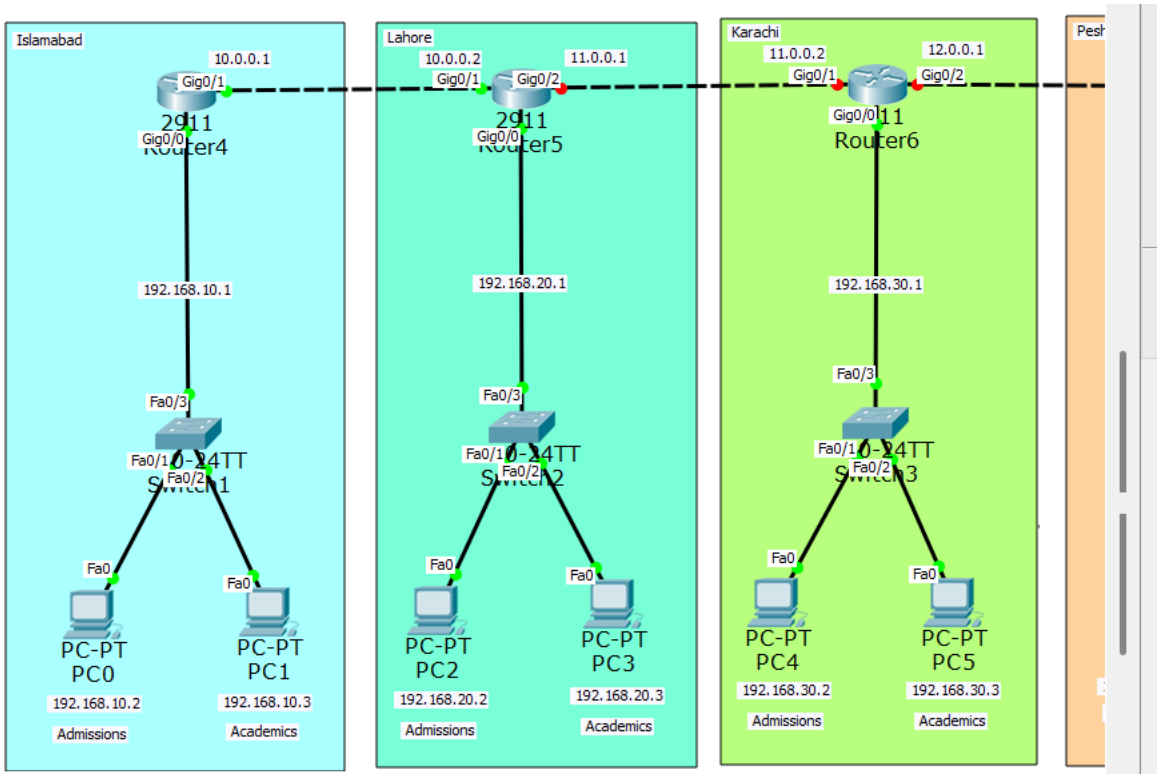
[INCOMING] MSG|FROM:Lahore|TO:Karachi|DEPT:Academics|BODY:Send the deaprtments list
menu> 1
Target Campus: Lahore
Department: Academics
Message: Sent via email
[SENT]
menu>
[INCOMING] MSG|FROM:Islamabad|TO:Karachi|DEPT:Adminstartion|BODY:Your amin member did not update us regarding summary.
menu> 1
Target Campus: Islamabad
Department: aminstartion
Message: NO message
[SENT]
menu> ^C
```

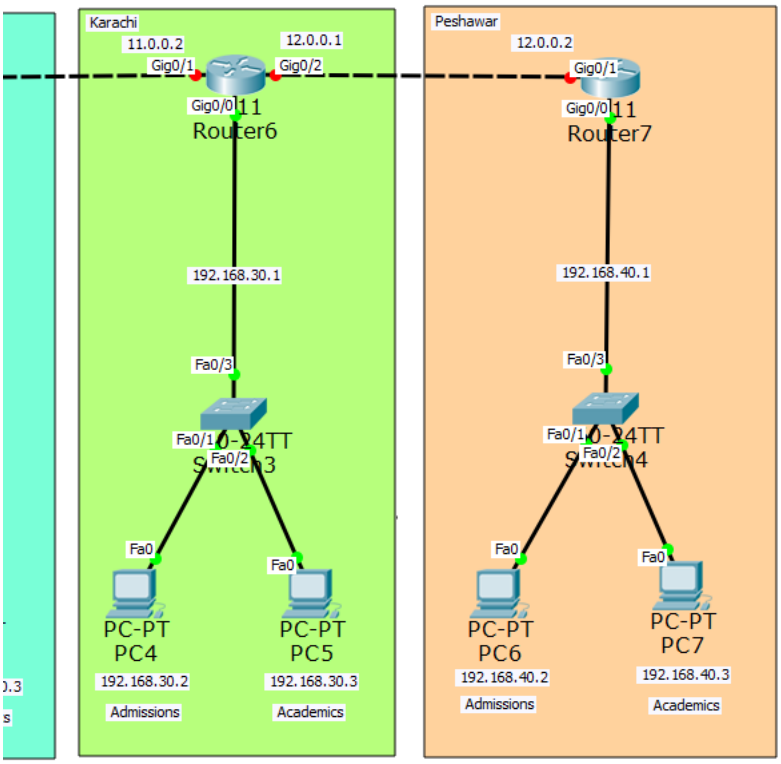
Islamabad campus:

```
maidah@ubuntu:~/Desktop/CN_LAB_Project$ g++ campus_client.cpp -pthread -o campus_client
maidah@ubuntu:~/Desktop/CN_LAB_Project$ ./campus_client 127.0.0.1 5000 5001 Islamabad 6003
[INFO] Connected to server.
Enter password for campus 'Islamabad': NU-ISB-123
[INFO] Authentication successful.
---- Campus Client: Islamabad ----
menu> 1
Target Campus: Karachi
Department: Adminstartion
Message: Your amin member did not update us regarding summary.
[SENT]
menu>
[INCOMING] MSG|FROM:Karachi|TO:Islamabad|DEPT:aminstartion|BODY:NO message
menu> ^C
```

Network Architecture – Part 2

a) Network Design





Configuring routers using rip version2:

Router4:

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.0
Router(config-router)#network 10.0.0.1
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
```

Router5:

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.20.0
Router(config-router)#network 10.0.0.2
Router(config-router)#network 11.0.0.1
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
```

Router6:

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.30.0
Router(config-router)#network 11.0.0.2
Router(config-router)#network 12.0.0.1
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
```

Router7:

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.40.0
Router(config-router)#network 12.0.0.2
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
```

b) IP Addressing Table:

Campus	PC1	PC2	Router(LAN)	Transit Link
Islamabad	**192.168.10.2**	**192.168.10.3**	**192.168.10.1**	**10.0.0.1**
Lahore	**192.168.20.2**	**192.168.20.3**	**192.168.20.1**	**10.0.0.2, 11.0.0.1**
Karachi	**192.168.30.2**	**192.168.30.3**	**192.168.30.1**	**11.0.0.2, 12.0.0.1**
Peshawar	**192.168.40.2**	**192.168.40.3**	**192.168.40.1**	**12.0.0.2**

c) Network Section: (RIPv2)

The network employs RIPv2 (Routing Information Protocol Version 2) as the sole dynamic routing protocol across the entire multi-campus WAN.

Routing Strategy

The use of RIPv2 provides an automated, hop-count based routing solution, essential for enabling end-to-end communication between all campus networks without manual static configuration.

- **Metric**: RIPv2 uses hop count (router count) to determine the best path, with a maximum limit of 15 hops.

- **Classless Support**: Version 2 is used specifically because it supports VLSM and CIDR. This is mandatory for using different subnet masks and prevents address wastage.
- **Configuration**: Key configurations on all routers include:
    - Enabling RIP Version 2.
    - Using the no auto-summary command to ensure correct advertisement of discontiguous and specific subnets.
    - Configuring LAN interfaces as passive interfaces to suppress unnecessary routing updates on campus local networks.

Students Detail:

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Semester Project Self-Evaluation (Fall 2025)

Name: Maidah Nasir	**Name: Anza Naseer**	**Name:**
Reg. No.: 23F-0764	**Reg. No.: 23F-0631**	**Reg. No.:**
Contribution to the Project's development: **Developed the following modules:** **1. UDP status monitoring** **2. TCP connection** **3. Packet Tracer Topology** **4. Report Preparation**	**Contribution to the Project's development:** **Developed the following modules:** **1. Message routing** **2. TCP/UDP communication** **3. Admin module** **4. Message send/receive**	**Contribution to the Project's development:** **Developed the following modules:** **1.** **2.** **3.** **4.**

Project completed (in %): _____100%_____
Project Modules Successfully Implemented:

1. **Central server**

2. **Campus Client**
3. **Admin and Network topology**
4. **Advance challenge dynamic routing**

Modules That Were Not Implemented:

1. **File transfer**
2. **Multi campus and department**
3.

Challenges faced & how they were overcome:

1. **Cisco throwed issues due to which lost my topology thrice as per it wasn't saving my file, registered a complaint.**
2.

Learnings from the Project:

1. Understanding how communication happens between different networks.

2. client - server communication via TCP/UDP, managing multiple concurrent connections.

3. Got to know how network works, message passing, IPs, LAN, WAN, routing, connectivity etc.

Comments (if any):

First ever lab which was completed with total understanding and learning, not using any tool for help yet the delivered concepts were clear enough for us to work. Thank you for the guidance throughout the sessions.
